CS7.503: Mobile Robotics — Mid Submission

Lifelong Mapping & 3D Change Detection on Object-Level Scene Graphs

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Code: https://github.com/PrajasW/LM-3DCD-SG

Abstract

A robot revisiting a room with a map from an earlier visit must (i) find its pose in that map without an initial guess, (ii) decide which objects stayed, moved, appeared or disappeared, and (iii) edit the map accordingly. We pose this as a joint relocalization and change-detection problem on object-level 3D scene graphs and propose a training-free pipeline whose core is a pairwise distance-consistency test: objects that did not move keep their mutual distances, so the largest mutually consistent set of object matches is the static set and yields the pose, while smaller consistent sets are objects that moved together. This report covers Phase 1. We formalize the problem, review the literature, implement graph construction on 3RScan/3DSSG and Steps 1–3 (candidate matching, maximum consistent set, pose), and evaluate relocalization against four baselines under one protocol in which the new visit is moved by an unknown rigid transform. On a 17-pair validation subset our method relocalizes 82.4% of pairs at the RIO threshold (20 cm/20°) in 0.13 s per pair, invariant to the transform. The learned scene-graph aligners SGAligner and SG-PGM fall to 35% and 18% (from 71–76% and 65% when frames are pre-aligned), while dense FPFH+RANSAC reaches 94% at 37× our runtime.

1 Problem statement

Map and observations. A map is an object-level 3D scene graph $\mathcal{G}^M = (\mathcal{V}^M, \mathcal{E}^M)$ in a fixed map frame. Each object node $o \in \mathcal{V}^M$ stores its points $P_o \subset \mathbb{R}^3$, centroid $\mathbf{c}_o$, extents $\mathbf{e}_o$, class ℓ_o, height above the floor h_o and change history; structural nodes hold floor and walls; edges carry support/spatial relations. A visit produces a point cloud split into segments, from which we build $\mathcal{G}^N = (\mathcal{V}^N, \mathcal{E}^N)$ in the robot's own frame, related to the map frame by an unknown $\mathbf{T}^\star \in \mathrm{SE}(3)$ (no initial guess, no odometry between visits).

Goal. Given $\mathcal{G}^M$, $\mathcal{G}^N$ and the visit's depth images, estimate

1. the pose $\hat{\mathbf{T}} \in \mathrm{SE}(3)$ such that $\hat{\mathbf{T}}\,\mathbf{c}_{o'} \approx \mathbf{c}_o$ for every unchanged object pair (o, o'), or report failure ($\perp$) when the observation cannot determine it;
2. a partition of the objects into *static* $S \subset \mathcal{V}^M \times \mathcal{V}^N$ (one-to-one matches), *moved* groups $\{(M_k, \mathbf{T}_k)\}$ with their rigid motions, *removed* $R \subset \mathcal{V}^M$, *added* $A \subset \mathcal{V}^N$ and *not observed* $U \subset \mathcal{V}^M$, where R and A require evidence of free space and everything else unexplained goes to U;
3. the updated map $\mathcal{G}^{M'}$ obtained by editing $\mathcal{G}^M$ per object, so that object identities persist and $|\mathcal{G}^{M'}|$ grows with the number of objects, not the number of visits.

Over a sequence of visits $t = 1, \ldots, K$ the map is the result of applying this update repeatedly.

Phase 1 sub-problem. With segments from 3RScan/3DSSG annotations, solve (1) and the static part of (2): return $(\hat{\mathbf{T}}, S)$ or $\perp$. A pair counts as relocalized if $\angle(\hat{\mathbf{R}}, \mathbf{R}^\star) \le \tau_r$ and $\|\hat{\mathbf{t}} - \mathbf{t}^\star\| \le \tau_t$ with the RIO thresholds $(\tau_t, \tau_r) \in \{(10\,\mathrm{cm}, 10°), (20\,\mathrm{cm}, 20°)\}$ [1]; $\perp$ counts as a failure. We also report registration recall (RMSE of the visit's points < 0.2 m, the criterion of [2–4]) and runtime.

2 Literature review

3D scene graphs. Armeni et al. [5] introduced layered 3D scene graphs; Kimera [6] and Hydra [7] build them online from SLAM, with objects assumed static and only agents moving. 3DSSG [8] annotates every 3RScan scan with an object-level graph (534 classes, 41 relations) and learns to predict it; SceneGraphFusion [9] predicts graphs incrementally from RGB-D, and ConceptGraphs [10] attaches open-vocabulary features to object nodes. Their update rules either assume a static world or delete objects that are not re-detected, irrespective of whether they were in view. We reuse this representation but make the per-visit edit explicit and evidence based.

Table 1: Capabilities of related systems (✓ yes, – no, ∘ partial). "Pose" = relocalization without initial alignment; "Changes" = per-object moved/added/removed; "Multi" = evaluated over > 2 visits.

Method	Pose	Changes	Map edit	Training-free	Multi
Kimera / Hydra [6, 7], ConceptGraphs [10]	–	–	∘	✓/✓/✓	–
SGAligner [2], SG-PGM [3], SG-Reg [4]	✓	–	–	–	–
ROMAN [16]	✓	–	–	✓	–
Objects Can Move [26], Living Scenes [27]	–	∘	–	✓/–	–
DSG [28]	–	∘ (no moved)	✓	∘ (LLM)	–
3D VSG [29]	–	predicted only	–	–	–
Ours (proposed)	✓	✓	✓	✓	✓

Scene-graph alignment and relocalization. RIO [1] introduced 3RScan and object instance re-localization across rescans. X-View [11] localizes with semantic graphs built from views. SGAligner [2] learns multi-modal node embeddings (point cloud, relations, attributes), matches the top-K nodes and registers the matched objects with GeoTransformer [12] and GC-RANSAC [13]. SG-PGM [3] casts alignment as partial graph matching, fuses point-level backbone features into nodes, and re-scores point correspondences with node matches. SG-Reg [4] targets generalization and efficiency with a lighter network and a robust back-end, SGAligner++ [14] adds language features, and OpenSGA [15] moves to open-world graphs. All of these are trained, report node matching and registration on sub-scans cut from the same scan, and ignore unmatched objects, so they do not output changes. ROMAN [16] aligns open-set object maps for global localization using CLIPPER [17], the closest training-free relative of our Step 2.

Consistency-based data association and robust registration. Joint compatibility [18] and PCM [19] select the largest set of mutually consistent measurements (PCM via a maximum clique for loop closures between maps); CLIPPER [17] relaxes it to a weighted dense subgraph; TEASER [20] and maximal-clique registration [21] apply the pairwise-invariant test to point correspondences. Classical global registration pairs FPFH features [22] with RANSAC [23] and refines by ICP [24, 25]. We lift the pairwise test from points to objects, which shrinks the problem from thousands of correspondences to tens and makes the consistent sets semantically meaningful (static set vs. objects moved together).

Change detection and long-term mapping. Objects Can Move [26] detects moved, added and removed parts between two aligned scans by geometric-transformation consistency, producing unlabelled clusters. Living Scenes [27] relocalizes and reconstructs multiple objects across 3RScan rescans, again given the scan alignment. DSG [28] labels objects stable/appeared/missing with Gaussian renderings and an LLM using ground-truth poses and has no "moved" state; 3D VSG [29] predicts change likelihood without verifying it. Panoptic Multi-TSDFs [30] and Khronos [31] maintain consistent object-level volumetric maps in changing scenes within one SLAM session, and LT-mapper [32] handles LiDAR lifelong mapping with change detection after alignment. Table 1 summarizes the gap: no method jointly relocalizes without an initial guess, labels per-object changes with free-space evidence and edits a persistent map over many visits without training.

3 Method and progress

Graph construction (done). We load the annotated instance PLY, semantic labels and 3DSSG relations of each 3RScan scan. Floor, wall and ceiling segments become structural nodes; every other segment with ≥ 50 points becomes an object node. A floor plane $(\mathbf{n}, d)$ is fitted to the floor segment and oriented towards the scene. Extents are pose invariant: the vertical extent along $\mathbf{n}$ and the two horizontal extents of the minimum-area bounding rectangle of the footprint (rotating calipers on its hull; PCA axes were unstable for square footprints, caught by a unit test), all as 2–98% percentile ranges. Edges are 3DSSG relations plus proximity edges (gap $< 0.3\,\mathrm{m}$). Figure 2 shows the two graphs of one pair.

Step 1: candidate matches (done). For map object a and visit object b of the same class,

$$\kappa(a,b) = \sum_{k=1}^{3} \left|\log(e_{a,k}+\epsilon) - \log(e_{b,k}+\epsilon)\right| + \frac{|h_a - h_b|}{\sigma_h}, \qquad \epsilon = 5\,\mathrm{cm},\ \sigma_h = 0.3\,\mathrm{m}, \tag{1}$$

and $\kappa = \infty$ across classes. Each map object keeps its $k = 3$ cheapest visit objects and vice versa (if $\kappa \leq 2.5$). Nothing depends on the robot's pose; identical objects deliberately receive several candidates, and the choice among them is left to geometry in Step 2.

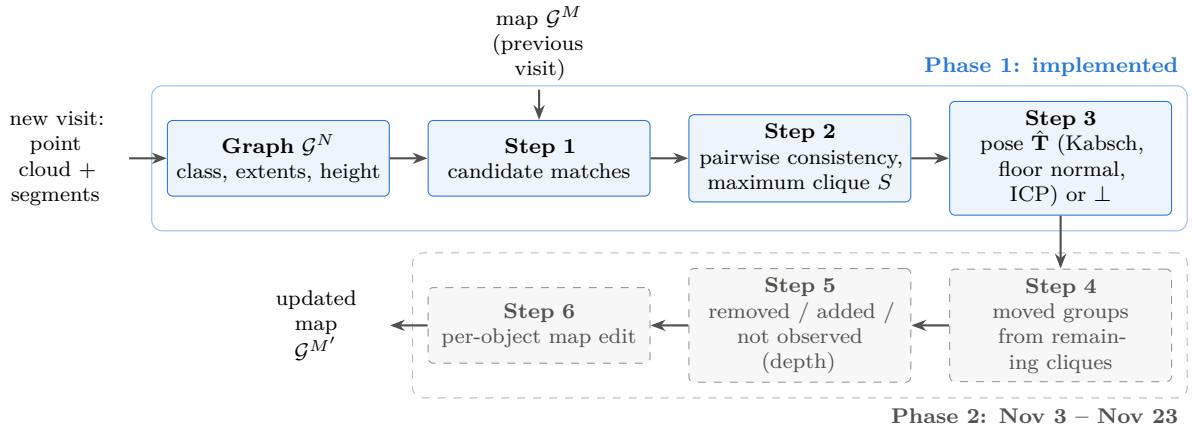


Figure 1: Pipeline for one visit. Phase 1 (blue) builds the visit's graph and relocalizes it in the map; the static set and the remaining consistent groups it produces are the inputs of Steps 4–6 (dashed, Phase 2).

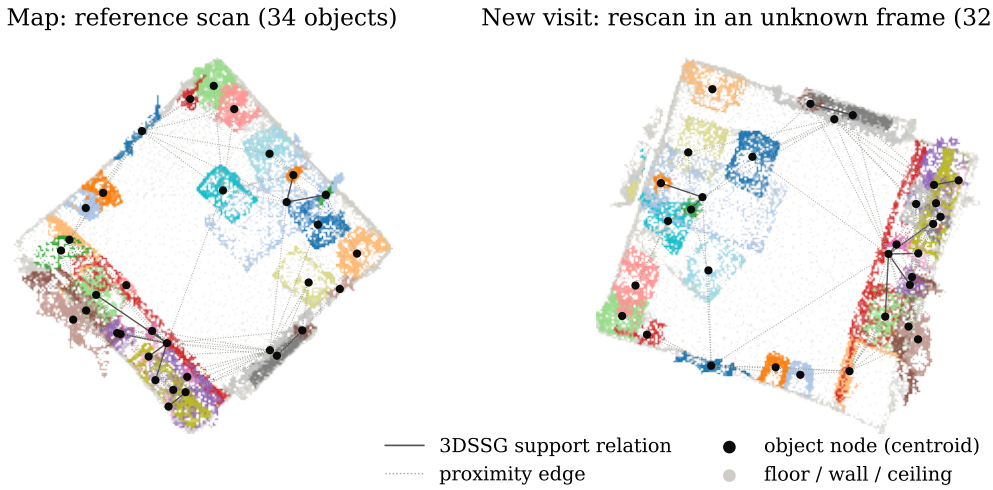


Figure 2: Object-level scene graphs built from the annotated segments of a 3RScan reference scan (left) and one rescan (right), top-down in each scan's own floor frame. Colours mark instances (shared IDs). The rescan is in an unknown frame; 3RScan annotates 11 of its objects as moved and one as removed. Edges: 3DSSG support relations and proximity.

Step 2: largest consistent set (done). Two candidates $m = (a, b)$ and $m' = (a', b')$ are compatible if they are one-to-one ($a \neq a'$, $b \neq b'$) and

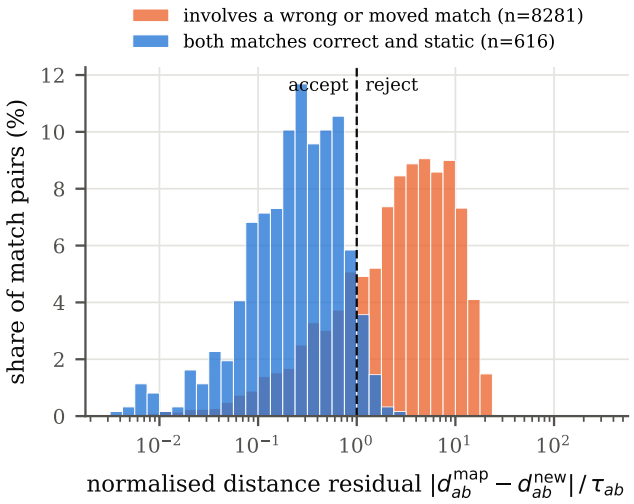
$$\big|\,\|\mathbf{c}_a - \mathbf{c}_{a'}\| - \|\mathbf{c}_b - \mathbf{c}_{b'}\|\,\big| \le \tau_{mm'} \quad \text{and} \quad \big|(h_a - h_{a'}) - (h_b - h_{b'})\big| \le \tau_{mm'}, \qquad \tau_{mm'} = \varepsilon + \beta\,(r_m + r_{m'}), \tag{2}$$

with $\varepsilon = 0.1\,\text{m}$ for sensor/annotation noise and $\beta = 0.1$ scaling with the object radii r because the centroid of a partly observed large object shifts. The height test rejects mirror-symmetric solutions that preserve all distances. The static set S is the maximum clique of the compatibility graph (igraph; ties broken by summed κ). Removing S's objects and re-solving yields further cliques of size ≥ 2, candidate groups of objects that moved together, which Step 4 will verify. Figure 3a shows the test on all candidate pairs of the validation subset: it keeps 94.5% of pairs between correct, unchanged matches and 25.6% of the rest; the clique removes the remainder, since a wrong match must be consistent with *every* member of S.

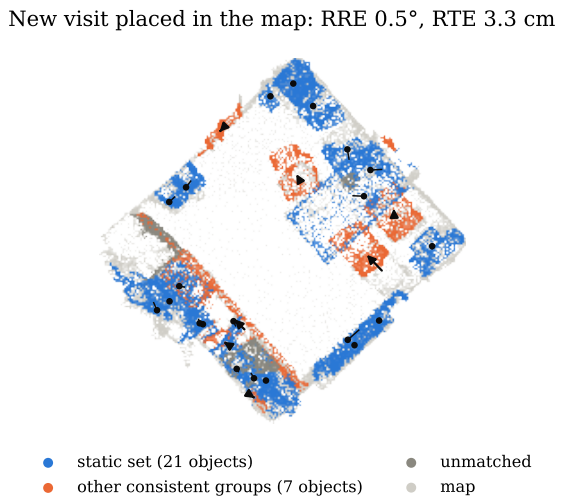
Step 3: pose (done). With $|S| \ge 3$ and non-collinear centroids (second singular value of the centred centroids $\ge 0.2\,\text{m}\cdot\sqrt{|S|}$) we solve

$$\hat{\mathbf{R}}, \hat{\mathbf{t}} = \arg\min_{\mathbf{R},\mathbf{t}} \sum_{(a,b)\in S} w_{ab}\|\mathbf{R}\mathbf{c}_b + \mathbf{t} - \mathbf{c}_a\|^2 + \lambda\,\|\mathbf{R}\,\mathbf{n}^N - \mathbf{n}^M\|^2, \qquad w_{ab} = (0.1 + \max(r_a, r_b))^{-1}, \tag{3}$$

in closed form (weighted Kabsch/Umeyama [33] with the floor normals as an extra direction pair), then refine with point-to-plane ICP [25] on the points of S and the structural nodes (5 cm voxels, correspondence radius $0.3 \to 0.1 \to 0.05\,\text{m}$). If $|S| < 3$, the centroids are collinear, or the ICP fitness is below 0.3, we return $\perp$. On the development rooms, allowing collinear sets (made solvable by the floor normal) only added wrong poses: the collinear sets were objects moved together, so the test stays.



(a) Pairwise test on all candidate pairs.



(b) Result for the pair of Fig. 2.

Figure 3: (a) Distance residual normalized by the tolerance $\tau_{mm'}$ for candidate pairs whose two matches are both correct and unchanged (blue) vs. all other pairs (orange), 17 validation pairs. (b) The visit placed in the map with the estimated pose (RTE 3.3 cm): the static clique (blue, black lines to map centroids) gives the pose. The remaining consistent groups (orange, arrows) are only candidates for Step 4: 4 of their 7 objects are annotated as moved (chairs, toaster); the others are static objects missed by S, which Step 4 must re-test against $\hat{\mathbf{T}}$.

Engineering. The code (`lifelong/`, ∼600 lines of Python) has unit tests on synthetic rooms (pose invariance, duplicate objects, a jointly moved group, declining). All thresholds were chosen on 6 *training* rooms (22 pairs); a sweep over ε, β, k and the cost gate gave 19/22 correct and 0 wrong poses almost everywhere, so results are not sensitive to them.

4 Experiments

Data and protocol. 4 validation rooms of 3RScan [1] with 3DSSG [8] (17 reference→rescan pairs; 11–34 objects per map, 6–32 per visit; 81 moved, 11 removed and 3 deformed instances in total). For each pair the rescan is moved by a stored random transform (uniform rotation in SE(3), translation $\mathcal{U}(-2, 2)^3$ m) and every method receives exactly these points; $\mathbf{T}^\star$ comes from the 3RScan alignment and is used only for scoring. A second run gives the rescan already aligned to the map, the setting of the learned baselines' training and evaluation.

Baselines. We selected the strongest methods with runnable 3RScan code: SG-PGM [3] and SGAligner [2] with their released 3RScan-trained checkpoints (rooms are from the validation split, so there is no train/test leakage); FPFH [22]+RANSAC and FPFH+TEASER++ [20] (training free, both with ICP). SG-Reg [4] (only a ScanNet-trained model and a C++ back-end released) and SGAligner++ [14] (no code) are cited from their papers. Our adapter exports the pairs into SGAligner's data layout and runs its unmodified preprocessing; SG-PGM reads the same files. Two fixes were needed for a fair run: GC-RANSAC in SGAligner requires correspondences sorted by score for its PROSAC sampler (unsorted input returned no model in 53% of aligned pairs), and FPFH normals are oriented towards the scan centre (Open3D's default sign depends on the frame, which halved TEASER++'s success under rotation).

Results. Table 2 and Fig. 4 give the comparison. (i) *Invariance.* Ours scores identically with and without the random transform. The learned aligners collapse (SGAligner 71–76%→35%, SG-PGM 65%→18%): their node encoders consume object points in world coordinates, and SG-PGM's published random-transform evaluation rotated only the registration points, not the scene-graph objects. (ii) *Accuracy and cost.* Ours matches FPFH+TEASER++ and trails FPFH+RANSAC by two pairs, at 37× lower cost: Steps 1–2 take 1.5 ms on 4–77 candidates, ICP the remaining 0.13 s. ICP halves the error of the centroid-only pose (7.6→3.3 cm) and raises succ.10/10 from 53% to 71%. Unlike the point-level methods, ours also returns the object correspondences (static-set precision 60.5%, recall 58.5% against unchanged instances; most "errors" are swaps among identical objects, or objects with small motions listed as changed, which do not hurt the pose). (iii) *Failures* (Fig. 5 in the appendix). One café room has a row of identical table–chair–menu sets: shifting by one set gives a consistent group of 8 objects against 7 for the true one, and with no walls in the scan ICP cannot tell; RANSAC also fails

Table 2: Two-visit relocalization on 17 validation pairs with a random rigid transform. "Ret." = pose returned; succ. = RIO thresholds; RR = RMSE < 0.2 m; median errors over returned poses; time per pair (pose only). Last column: succ.20/20 when the rescan is pre-aligned (the baselines' own setting).

Method	Ret. (%)	succ. 10/10	succ. 20/20	RR	RRE (°)	RTE (cm)	Time (s)	Aligned 20/20
SGAligner [2]	47.1	35.3	35.3	35.3	1.24	4.7	2.73	70.6 – 76.5
SG-PGM [3]	100	17.6	17.6	17.6	80.2	145	2.99	64.7
FPFH + TEASER++ [20]	100	70.6	82.4	82.4	1.07	3.2	1.69	82.4
FPFH + RANSAC [22]	100	**76.5**	**94.1**	**94.1**	1.07	**3.2**	4.82	88.2
Ours, centroid pose only	94.1	52.9	76.5	82.4	1.55	7.6	<0.01	82.4
Ours (Steps 1–3)	94.1	70.6	82.4	82.4	**0.99**	3.3	**0.13**	82.4

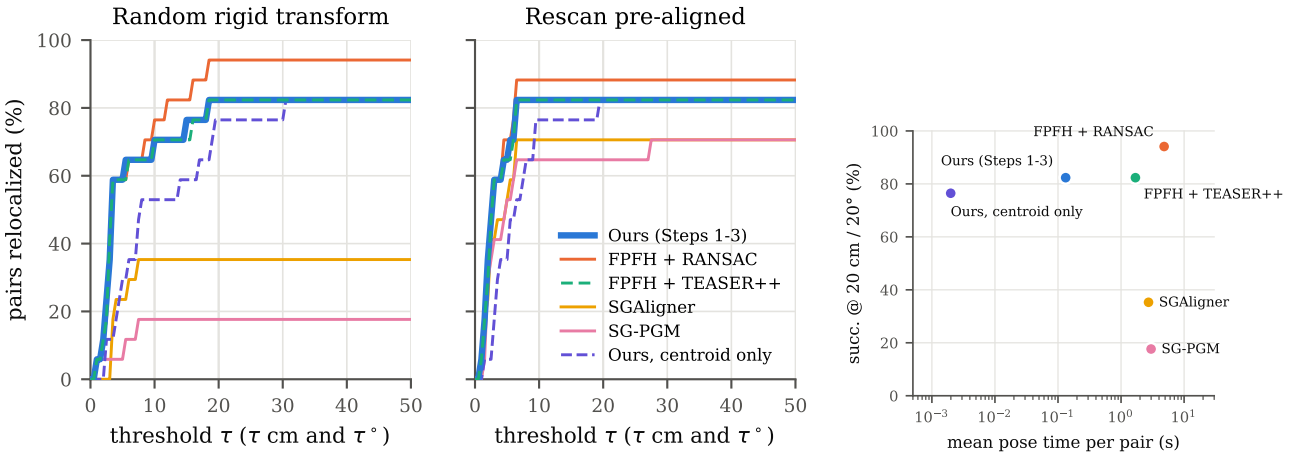


Figure 4: Left: share of pairs relocalized as the threshold grows (translation τ cm and rotation $\tau°$ jointly), with a random transform (our protocol) and with pre-aligned frames. Ours and FPFH+TEASER++ coincide on this subset except near 15 cm. Right: success at 20 cm/20° against mean pose time.

one of its two pairs. One rescan sees only 6 objects and we decline (RANSAC succeeds on walls and floor). On the development rooms the remaining declines are rooms in which *every* object moved, for which declining is the correct object-level answer, but where structure-only registration would still give the pose.

5 Timeline for the remaining work

Table 3: Plan for Phase 2 and the demo (weekly milestones).

Dates	Work package	Deliverable
Oct 27 – Nov 2	Full 3RScan validation split (47 rooms, 110 rescans) for all methods; structure-only fallback (floor/wall registration) when $\|S\| < 3$; repeated-layout check using wall evidence.	Final Phase-1 table
Nov 3 – Nov 9	**Step 4**: motion of each remaining clique (Kabsch + ICP), single-object moves by direct ICP gated by Step 5; **Step 5**: project unmatched map objects into the visit's depth frames (`sequence.zip`) with $\hat{\mathbf{T}}$; free space ⇒ removed, occluded/out of view ⇒ not observed; added only over previously observed free space.	Change P/R per type
Nov 10 – Nov 16	**Step 6**: per-object map edit with change history and free-space record; multi-visit setting (rooms with ≥ 3 scans, chronological order); map size and "correct place" share after each visit.	Multi-visit curves
Nov 17 – Nov 23	Predicted segments (SceneGraphFusion-style or class-agnostic segments; class mismatch cost instead of hard gate); ablations; final report.	Final report (Nov 23)
Nov 24 – Nov 26	Demo: relocalization, change labels and map edits on 3RScan rooms (interactive viewer + video).	Demo (Nov 26)

Risks. Predicted segments break the one-to-one object assumption (over/under-segmentation); we will allow many-to-one candidates and rely on the clique. Step 5 needs the RGB-D sequences (∼80 MB per scan); we will subsample frames. The evaluation subset is small (one pair ≈ 6 points), so the full split comes first.

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A Appendix: per-pair results and implementation details

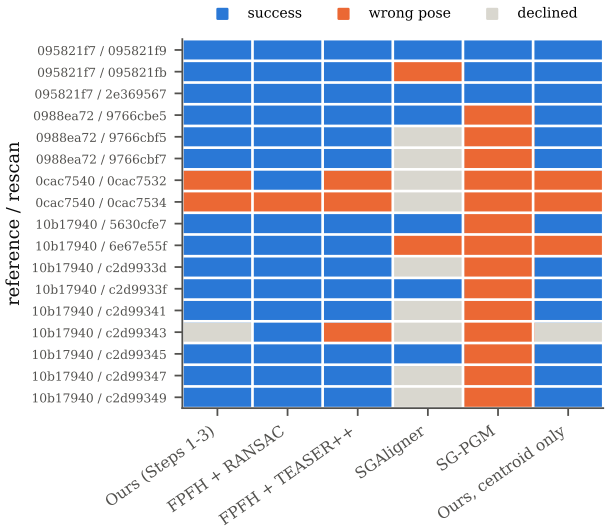


Figure 5: Per-pair outcome at 20 cm/20° (random transform). Rows are grouped by reference scan.

Table 4: Parameters of our method (`configs/default.yaml`); chosen on 6 training rooms.

Step	Parameter	Value
Graph	min. segment size	50 points
1	candidates per object k	3 (both directions)
1	extent offset ϵ, height scale σ_h	5 cm, 0.3 m
1	cost gate	$\kappa \leq 2.5$, same class
2	tolerance ε, β	0.10 m, 0.10
2	moved-group min. size	2
3	min. static objects / spread	3 / 0.2 m
3	ICP voxel, radii	5 cm; 0.3, 0.1, 0.05 m
3	min. ICP fitness	0.3

Baseline setup. One conda environment (PyTorch 2.7.1/CUDA 12.8, RTX 2080 Ti) runs SGAligner (K=2 node matches, 3DMatch GeoTransformer, GC-RANSAC) and SG-PGM (released weights, GeoTransformer backbone). SGAligner's own preprocessing builds the per-scan pickles from the exported full scans (attribute vocabulary 164, relations 41, as in the checkpoints); 3DSSG "inside" relations are dropped because that preprocessing rejects them. SG-PGM's internal augmentation is disabled since the transform is already in the data. Source commits, patches and exact package versions are in the repository (`baselines/`).

Reproduction. `bash scripts/run_all.sh` runs tests, pairs, all methods and evaluation; `scripts/make_figures.py` regenerates every figure in this report from `results/`.